

Workshop Job Card MVP - Database Schema Document

This document defines the business-ready database structure for the single-tenant workshop application. It is based on the approved PRD, revised wireframes, and your confirmed workshop rules.

1. Purpose of this schema

The database must support:

- linked customer and vehicle records
- ownership history
- multiple job cards per vehicle
- multiple complaints per job card
- multiple mechanics per job card
- complaint capture via icons, text, audio, video, and photos
- parts requests, ordering, receiving, dispatch, and usage
- manager pricing control
- admin import and merge from legacy software
- workshop profile and operational settings
- future Zoho integration
- future multi-tenant conversion without redesign

This is a ****single-tenant MVP schema****, but it is designed so a future ``tenant_id`` can be added cleanly.

2. Core business rules confirmed

Customer and vehicle

- One customer can have multiple vehicles.
- One vehicle has one current active owner at a time.
- One vehicle can change owner later.
- Ownership history must be preserved.
- Customer matching should use a ****combination of identifiers**** like mobile, GST, email, and name.

Vehicle identity

- Primary unique identity = ****registration number****.
- Registration numbers should be normalized to a standard format.
- VIN/chassis remains an optional secondary identity when available.

Job card rules

- One vehicle can have multiple job cards.
- One job card can have multiple complaints.
- One job card can have multiple mechanics.
- Closed job cards can be reopened.
- Imported historical job cards remain read-only by default.

Complaint capture

- Complaints can be captured by:
 - icon selection
 - text
 - audio recording
 - video recording
 - image/photo attachment
- At least one complaint method is required before final job card creation.
- Separate fields are required for:
 - customer complaint
 - advisor observation
 - internal technical note

Parts workflow

- Mechanics can request parts.
- Parts Manager can create supplier orders and dispatch parts.
- Manager controls commercial approval and selling price.
- Parts statuses are confirmed as:
 - Requested
 - Approved
 - In Stock
 - Ordered
 - Partially Received
 - Dispatched
 - Used
 - Rejected
 - Cancelled
- Supplier records should be stored from day one.

Labour workflow

- Mechanics can add labour tasks and notes.
- Manager approves labour pricing.
- Time tracking is not required for MVP.

Media

- Media types to store:
 - photos
 - videos
 - audio recordings
 - signatures
 - diagnostics files
- Delivery photos are required when available.
- Primary media link should be the ****job card****.

Import and merge

- Both one-time migration and repeated imports must be supported.
- Import templates should be saved by source software.
- Duplicate merge approval can be done by ****Admin and Manager****.
- Unknown import columns should be stored temporarily, not discarded.

Roles

- MVP roles:
 - Admin
 - Manager
 - Advisor
 - Mechanic
 - Parts Manager
 - Accounts
- Multiple roles per user are allowed.

Workshop profile

- Full workshop profile setup is included.

Billing and tax

- Store ****part selling price****.
- Store ****labour selling price****.
- Tax model:
 - intra-state = CGST 9% + SGST 9%
 - inter-state = IGST 18%
- Discounts supported at both line-item and overall job level.

Service history and reminders

- Vehicle history should retain:
 - previous job cards
 - repeated complaints
 - previous odometer
 - previous parts used
 - diagnostics history
- Reminder fields should be stored where data is available for:
 - next service
 - next oil change
 - next PUC
 - insurance expiry

3. Database design approach

The schema is grouped into these layers:

1. ****Setup and control tables****
2. ****Master tables****
3. ****Transaction tables****
4. ****Media and attachment tables****
5. ****Parts procurement tables****
6. ****Import and merge tables****
7. ****Audit and system tables****

4. Setup and control tables

4.1 workshop_profile

****Purpose:**** stores the full workshop identity and branding profile.

Field	Type	Required	Notes
id	UUID / bigint	Yes	Primary key
workshop_name	text	Yes	Legal or operating name
brand_name	text	No	Display brand
logo_url	text	No	Stored file link
address_line_1	text	Yes	
address_line_2	text	No	
city	text	No	
state	text	No	Needed for GST/place of supply
postal_code	text	No	
country	text	No	Default IN
mobile	text	No	Main phone
email	text	No	
website	text	No	
gst_in	text	No	
pan	text	No	
workshop_timings	text	No	Free text for MVP
invoice_footer_text	text	No	
terms_conditions_text	text	No	
whatsapp_template_text	text	No	
created_at	datetime	Yes	
updated_at	datetime	Yes	

4.2 tax_settings

****Purpose:**** default tax behavior for pricing and billing.

Field	Type	Required	Notes
id	UUID / bigint	Yes	Primary key
workshop_profile_id	FK	Yes	One active config for MVP
default_tax_mode	enum	Yes	exclusive / inclusive
intrastate_cgst_rate	decimal(5,2)	Yes	default 9.00
intrastate_sgst_rate	decimal(5,2)	Yes	default 9.00
interstate_igst_rate	decimal(5,2)	Yes	default 18.00
default_discount_mode	enum	Yes	line_item / overall / both
created_at	datetime	Yes	
updated_at	datetime	Yes	

4.3 numbering_settings

****Purpose:**** stores numbering formats.

Field	Type	Required	Notes
id	UUID / bigint	Yes	
workshop_profile_id	FK	Yes	
estimate_number_format	text	Yes	Example EST-{YYYY}-{####}
jobcard_number_format	text	Yes	Example JC-{YYYY}-{####}
parts_order_number_format	text	Yes	Example PO-{YYYY}-{####}
created_at	datetime	Yes	
updated_at	datetime	Yes	

4.4 feature_flags

****Purpose:**** enable/disable features without code changes.

Field	Type	Required	Notes
id	UUID / bigint	Yes	
feature_key	text	Yes	unique
feature_name	text	Yes	display name
is_enabled	boolean	Yes	
config_json	json/jsonb	No	optional extra settings
updated_at	datetime	Yes	

Examples:

- plate_scan
- intake_360_photos
- intake_video
- audio_complaints
- customer_signature
- whatsapp_share
- diagnostics_upload
- offline_sync
- ownership_history
- duplicate_warning

4.5 workflow_settings

****Purpose:**** controls workflow validations and locks.

Field	Type	Required	Notes
id	UUID / bigint	Yes	

id	UUID / bigint	Yes	
mandatory_odometer	boolean	Yes	
mandatory_signature	boolean	Yes	
mandatory_photos	boolean	Yes	
minimum_intake_photo_count	integer	Yes	
manager_approval_before_print	boolean	Yes	
lock_after_close	boolean	Yes	
allow_reopen_closed_job	boolean	Yes	
duplicate_vehicle_action	enum	Yes	warn / block
duplicate_customer_action	enum	Yes	warn / block
imported_jobs_read_only	boolean	Yes	
updated_at	datetime	Yes	

4.6 print_settings

****Purpose:**** controls PDF/print defaults.

Field	Type	Required	Notes
---	---	---:	---
id	UUID / bigint	Yes	
workshop_profile_id	FK	Yes	
show_tax_by_default	boolean	Yes	
show_discount_by_default	boolean	Yes	
include_signature	boolean	Yes	
include_intake_photos	boolean	Yes	
show_parts_labour_separately	boolean	Yes	
updated_at	datetime	Yes	

5. User, role, and permission tables

5.1 users

****Purpose:**** stores all system users.

Field	Type	Required	Notes
---	---	---:	---
id	UUID / bigint	Yes	
full_name	text	Yes	
mobile	text	No	login/contact
email	text	No	
password_hash	text	No	if password auth used
quick_pin_hash	text	No	optional for workshop devices
profile_photo_url	text	No	
skill_category	text	No	useful for mechanics
is_active	boolean	Yes	
last_login_at	datetime	No	

created_at	datetime	Yes	
updated_at	datetime	Yes	

5.2 roles

****Purpose:**** defines system roles.

Field	Type	Required	Notes	
---	---	---	---	
id	UUID / bigint	Yes		
role_key	text	Yes	unique	
role_name	text	Yes		
description	text	No		

Seed roles:

- admin
- manager
- advisor
- mechanic
- parts_manager
- accounts

5.3 user_roles

****Purpose:**** supports multiple roles per user.

Field	Type	Required	Notes	
---	---	---	---	
id	UUID / bigint	Yes		
user_id	FK	Yes		
role_id	FK	Yes		
is_primary	boolean	Yes	exactly one primary role recommended	
created_at	datetime	Yes		

5.4 permissions

****Purpose:**** permission catalog.

Field	Type	Required	Notes	
---	---	---	---	
id	UUID / bigint	Yes		
permission_key	text	Yes	unique	
permission_name	text	Yes		
module_name	text	No		

description	text	No	
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5.5 role_permissions

****Purpose:**** maps permissions to roles.

Field	Type	Required	Notes	
---	---	---	---	
id	UUID / bigint	Yes		
role_id	FK	Yes		
permission_id	FK	Yes		
is_allowed	boolean	Yes		

6. Master tables

6.1 customers

****Purpose:**** stores customer master data.

Field	Type	Required	Notes	
---	---	---	---	
id	UUID / bigint	Yes		
customer_code	text	No	internal code	
customer_type	enum/text	Yes	retail / corporate / insurance / dealer	
display_name	text	Yes		
billing_name	text	No		
primary_mobile	text	No	strong matching field	
alternate_mobile	text	No		
email	text	No		
gst_number	text	No		
address_line_1	text	No		
address_line_2	text	No		
city	text	No		
state	text	No	useful for GST logic	
postal_code	text	No		
country	text	No		
notes	text	No		
is_active	boolean	Yes	no hard delete	
source_system	text	No	import reference	
source_record_id	text	No	legacy id	
created_at	datetime	Yes		
updated_at	datetime	Yes		

****Indexes recommended:****

- primary_mobile
- gst_number

- email
- display_name

6.2 vehicles

****Purpose:**** stores vehicle master data.

Field	Type	Required	Notes
id	UUID / bigint	Yes	
vehicle_code	text	No	internal code
registration_number_raw	text	Yes	original input
registration_number_normalized	text	Yes	unique primary identity
vin	text	No	optional secondary identity
engine_number	text	No	
manufacturer	text	No	
model	text	No	
variant	text	No	
fuel_type	text	No	
color	text	No	
manufacture_year	integer	No	
battery_details	text	No	
current_odometer	integer	No	last known reading
insurer_name	text	No	
insurance_expiry_date	date	No	reminder if available
next_service_date	date	No	
next_puc_date	date	No	
next_oil_change_date	date	No	
next_oil_change_distance	integer	No	
next_timing_belt_change_date	date	No	
notes	text	No	
current_customer_id	FK	Yes	active owner
is_active	boolean	Yes	
source_system	text	No	import reference
source_record_id	text	No	legacy id
created_at	datetime	Yes	
updated_at	datetime	Yes	

****Constraints:****

- registration_number_normalized should be unique for active records.

6.3 vehicle_ownership_history

****Purpose:**** tracks ownership changes over time.

Field	Type	Required	Notes
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Field	Type	Required	Notes
id	UUID / bigint	Yes	
vehicle_id	FK	Yes	
customer_id	FK	Yes	
from_date	date	No	
to_date	date	No	null when current
transfer_notes	text	No	
created_at	datetime	Yes	

6.4 suppliers

Purpose: stores supplier master data for parts ordering.

Field	Type	Required	Notes
id	UUID / bigint	Yes	
supplier_name	text	Yes	
contact_person	text	No	
mobile	text	No	
email	text	No	
gst_number	text	No	
address_line_1	text	No	
address_line_2	text	No	
city	text	No	
state	text	No	
notes	text	No	
is_active	boolean	Yes	
created_at	datetime	Yes	
updated_at	datetime	Yes	

6.5 parts_master

Purpose: stores parts/items catalog.

Field	Type	Required	Notes
id	UUID / bigint	Yes	
part_name	text	Yes	
item_code	text	No	
sku	text	No	
part_number	text	No	
barcode	text	No	
brand	text	No	
unit	text	No	pcs, set, etc.
default_tax_rate	decimal(5,2)	No	
default_selling_price	decimal(12,2)	No	per your decision

stock_quantity	decimal(12,2)	No	optional simple stock field for MVP
is_active	boolean	Yes	
source_system	text	No	
source_record_id	text	No	
created_at	datetime	Yes	
updated_at	datetime	Yes	

6.6 labour_master

****Purpose:**** stores labour/service catalog.

Field	Type	Required	Notes
id	UUID / bigint	Yes	
labour_name	text	Yes	
labour_code	text	No	
unit_type	text	No	job / hour / flat
default_tax_rate	decimal(5,2)	No	
default_selling_price	decimal(12,2)	No	per your decision
standard_description	text	No	
is_active	boolean	Yes	
source_system	text	No	
source_record_id	text	No	
created_at	datetime	Yes	
updated_at	datetime	Yes	

6.7 complaint_icon_master

****Purpose:**** stores selectable complaint categories/icons.

Field	Type	Required	Notes
id	UUID / bigint	Yes	
icon_key	text	Yes	unique
display_name	text	Yes	
category	text	No	
sort_order	integer	No	
is_active	boolean	Yes	
created_at	datetime	Yes	

7. Transaction tables

7.1 jobcards

****Purpose:**** main service transaction record.

Field	Type	Required	Notes
id	UUID / bigint	Yes	
jobcard_number	text	Yes	unique
customer_id	FK	Yes	current linked customer
vehicle_id	FK	Yes	
advisor_id	FK users	No	creator/front desk
primary_mechanic_id	FK users	No	optional main mechanic
status	enum	Yes	Draft/Open/In Progress/etc.
date_in	datetime	Yes	
expected_delivery_at	datetime	No	
closed_at	datetime	No	
intake_odometer	integer	No	
fuel_level	text	No	
external_notes	text	No	customer-visible notes
internal_notes	text	No	internal remarks
customer_signature_required	boolean	Yes	snapshot of workflow rule
intake_photos_required	boolean	Yes	snapshot of workflow rule
legacy_import_flag	boolean	Yes	
read_only_flag	boolean	Yes	imported history default true
overall_discount_type	enum	No	amount / percent
overall_discount_value	decimal(12,2)	No	
subtotal_amount	decimal(12,2)	No	calculated
tax_amount	decimal(12,2)	No	calculated
total_amount	decimal(12,2)	No	calculated
place_of_supply_state	text	No	needed for tax mode
payment_status	enum	No	unpaid/partial/paid
zoho_sync_status	text	No	future use
source_system	text	No	import trace
source_record_id	text	No	legacy job id
created_at	datetime	Yes	
updated_at	datetime	Yes	

7.2 jobcard_snapshots

****Purpose:**** immutable copy of customer/vehicle data at time of job creation.

Field	Type	Required	Notes
id	UUID / bigint	Yes	
jobcard_id	FK	Yes	one-to-one
customer_name	text	Yes	
customer_mobile	text	No	
customer_address	text	No	
customer_gst_number	text	No	
vehicle_registration_number	text	Yes	

vehicle_manufacturer	text	No	
vehicle_model	text	No	
vehicle_color	text	No	
vehicle_vin	text	No	
intake_odometer_snapshot	integer	No	
created_at	datetime	Yes	

7.3 jobcard_mechanics

****Purpose:**** supports multiple mechanics per job card.

Field	Type	Required	Notes
id	UUID / bigint	Yes	
jobcard_id	FK	Yes	
mechanic_user_id	FK users	Yes	
assignment_role	text	No	electrical / AC / body etc.
is_primary	boolean	Yes	
assigned_at	datetime	Yes	
completed_at	datetime	No	

7.4 jobcard_complaints

****Purpose:**** stores structured complaint records per job card.

Field	Type	Required	Notes
id	UUID / bigint	Yes	
jobcard_id	FK	Yes	
customer_complaint_text	text	No	
advisor_observation_text	text	No	
internal_technical_note	text	No	
has_icon_input	boolean	Yes	
has_text_input	boolean	Yes	
has_audio_input	boolean	Yes	
has_video_input	boolean	Yes	
has_photo_input	boolean	Yes	
created_by_user_id	FK users	No	
created_at	datetime	Yes	
updated_at	datetime	Yes	

****Validation rule:****

At least one of the five input modes must be true before final job creation.

7.5 jobcard_complaint_icons

****Purpose:**** many-to-many mapping between complaints and selected icons.

Field	Type	Required	Notes
id	UUID / bigint	Yes	
jobcard_complaint_id	FK	Yes	
complaint_icon_id	FK	Yes	

7.6 jobcard_labour

****Purpose:**** labour tasks/lines on a job card.

Field	Type	Required	Notes
id	UUID / bigint	Yes	
jobcard_id	FK	Yes	
labour_master_id	FK	No	optional if chosen from master
labour_name	text	Yes	copied for stability
added_by_user_id	FK users	No	mechanic/manager
assigned_mechanic_user_id	FK users	No	
status	enum	Yes	pending/in_progress/completed/approved
mechanic_note	text	No	
selling_price	decimal(12,2)	No	manager-controlled
tax_rate	decimal(5,2)	No	
discount_type	enum	No	amount / percent
discount_value	decimal(12,2)	No	
quantity	decimal(12,2)	No	default 1
created_at	datetime	Yes	
updated_at	datetime	Yes	

7.7 jobcard_parts

****Purpose:**** parts lines on a job card.

Field	Type	Required	Notes
id	UUID / bigint	Yes	
jobcard_id	FK	Yes	
part_master_id	FK	No	optional if matched
part_name	text	Yes	copied for stability
item_code	text	No	
part_number	text	No	
brand	text	No	
requested_by_user_id	FK users	No	usually mechanic

approved_by_user_id	FK users	No	manager	
parts_manager_user_id	FK users	No	workflow owner	
quantity_requested	decimal(12,2)	Yes		
quantity_received	decimal(12,2)	No		
quantity_dispatched	decimal(12,2)	No		
quantity_used	decimal(12,2)	No		
status	enum	Yes	requested/approved/in_stock/etc.	
mechanic_note	text	No		
dispatch_note	text	No		
selling_price	decimal(12,2)	No	manager-controlled	
tax_rate	decimal(5,2)	No		
discount_type	enum	No	amount / percent	
discount_value	decimal(12,2)	No		
created_at	datetime	Yes		
updated_at	datetime	Yes		

7.8 jobcard_status_history

****Purpose:**** status audit trail for each job card.

Field	Type	Required	Notes	
---	---	---	---	
id	UUID / bigint	Yes		
jobcard_id	FK	Yes		
old_status	text	No		
new_status	text	Yes		
changed_by_user_id	FK users	No		
change_note	text	No		
changed_at	datetime	Yes		

8. Media and attachment tables

8.1 jobcard_media

****Purpose:**** unified storage for job card-linked media and files.

Field	Type	Required	Notes	
---	---	---	---	
id	UUID / bigint	Yes		
jobcard_id	FK	Yes	primary link	
complaint_id	FK	No	if complaint-specific	
labour_id	FK	No	if labour-specific	
part_id	FK	No	if part-specific	
media_type	enum	Yes	intake_photo / intake_video / complaint_audio / complaint_video / complaint_photo / delivery_photo / work_photo / signature / diagnostics_file / other	

phase	enum	No	intake / work / delivery / diagnostics
file_url	text	Yes	stored object URL
file_name	text	No	
mime_type	text	No	
file_size_bytes	bigint	No	
capture_label	text	No	front, rear, dashboard, etc.
captured_by_user_id	FK users	No	
captured_at	datetime	No	
created_at	datetime	Yes	

8.2 diagnostics_reports

****Purpose:**** explicit index for diagnostics records.

Field	Type	Required	Notes
---	---	---:	---
id	UUID / bigint	Yes	
jobcard_id	FK	No	can link to a job
vehicle_id	FK	Yes	history always relevant
media_id	FK jobcard_media	No	underlying uploaded file
source_type	enum	No	email / bluetooth / manual_upload
source_reference	text	No	email subject / device name / etc.
notes	text	No	
created_at	datetime	Yes	

9. Parts procurement and dispatch tables

9.1 supplier_orders

****Purpose:**** purchase order header for parts procurement.

Field	Type	Required	Notes
---	---	---:	---
id	UUID / bigint	Yes	
order_number	text	Yes	unique
supplier_id	FK	Yes	
created_by_user_id	FK users	No	usually parts manager
status	enum	Yes	draft/ordered/partial/received/cancelled
expected_arrival_date	date	No	
notes	text	No	
created_at	datetime	Yes	
updated_at	datetime	Yes	

9.2 supplier_order_lines

****Purpose:**** line items within supplier orders.

Field	Type	Required	Notes
id	UUID / bigint	Yes	
supplier_order_id	FK	Yes	
jobcard_part_id	FK	No	line may fulfill a request
part_master_id	FK	No	
part_name	text	Yes	
quantity_ordered	decimal(12,2)	Yes	
quantity_received	decimal(12,2)	No	
status	enum	Yes	ordered/partial/received/cancelled
notes	text	No	
created_at	datetime	Yes	
updated_at	datetime	Yes	

9.3 parts_dispatch_logs

****Purpose:**** records dispatch of parts to a job/mechanic.

Field	Type	Required	Notes
id	UUID / bigint	Yes	
jobcard_part_id	FK	Yes	
dispatched_by_user_id	FK users	No	parts manager
dispatched_to_user_id	FK users	No	mechanic
quantity_dispatched	decimal(12,2)	Yes	
dispatch_note	text	No	
dispatched_at	datetime	Yes	

10. Import and merge tables

10.1 import_profiles

****Purpose:**** saved import templates by source software.

Field	Type	Required	Notes
id	UUID / bigint	Yes	
profile_name	text	Yes	
source_software_name	text	No	
supported_file_types	text/json	No	csv/xlsx/etc.
header_detection_mode	text	No	auto/manual
grouping_mode	text	No	rows/columns/tabs
normalization_rules_json	json/jsonb	No	plate/phone/date rules

mapping_template_json	json/jsonb	No	saved field mappings
merge_rules_json	json/jsonb	No	saved duplicate logic
created_by_user_id	FK users	No	
created_at	datetime	Yes	
updated_at	datetime	Yes	

10.2 import_batches

****Purpose:**** one record per import operation.

Field	Type	Required	Notes
id	UUID / bigint	Yes	
batch_code	text	Yes	human-readable batch id
import_profile_id	FK	No	
import_type	text	Yes	customers/vehicles/jobcards/etc.
source_file_name	text	Yes	
import_mode	enum	Yes	dry_run/create/update/create_update/merge_review
total_rows	integer	No	
valid_rows	integer	No	
created_count	integer	No	
updated_count	integer	No	
merged_count	integer	No	
skipped_count	integer	No	
failed_count	integer	No	
exact_match_count	integer	No	
likely_match_count	integer	No	
unknown_field_count	integer	No	
status	enum	Yes	uploaded/validated/imported/failed
imported_by_user_id	FK users	No	
created_at	datetime	Yes	
updated_at	datetime	Yes	

10.3 import_batch_rows

****Purpose:**** row-level import result log.

Field	Type	Required	Notes
id	UUID / bigint	Yes	
import_batch_id	FK	Yes	
row_number	integer	Yes	
source_key	text	No	e.g. registration or customer id
detected_entity_type	text	No	vehicle/customer/etc.
row_status	enum	Yes	created/updated/merged/skipped/failed/pending_review
confidence_score	decimal(5,2)	No	optional

matched_record_id	text	No	
error_message	text	No	
unknown_fields_json	json/jsonb	No	temp storage for unmapped columns
raw_row_json	json/jsonb	No	original row snapshot
created_at	datetime	Yes	

10.4 duplicate_review_queue

****Purpose:**** stores unresolved likely duplicate cases.

Field	Type	Required	Notes	
---	---	---	:	---
id	UUID / bigint	Yes		
import_batch_row_id	FK	Yes		
entity_type	text	Yes	customer/vehicle/part/labour	
candidate_existing_record_id	text	No		
confidence_score	decimal(5,2)	No		
review_status	enum	Yes	pending/approved/merge/keep_separate/rejected	
reviewed_by_user_id	FK users	No	admin or manager	
review_note	text	No		
reviewed_at	datetime	No		
created_at	datetime	Yes		

10.5 merge_logs

****Purpose:**** permanent log of merge actions.

Field	Type	Required	Notes	
---	---	---	:	---
id	UUID / bigint	Yes		
entity_type	text	Yes	customer/vehicle/part/labour	
source_record_id	text	Yes	imported/duplicate record	
target_record_id	text	Yes	surviving master record	
merge_action_json	json/jsonb	No	field-by-field choices	
merged_by_user_id	FK users	No	admin or manager	
merged_at	datetime	Yes		

11. Audit and system tables

11.1 audit_logs

****Purpose:**** generic audit trail for important business actions.

Field	Type	Required	Notes	
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Field	Type	Required	Notes
id	UUID / bigint	Yes	
user_id	FK users	No	
module_name	text	Yes	jobcard/import/settings/etc.
action_key	text	Yes	create/update/delete/merge/close/reopen
entity_type	text	Yes	
entity_id	text	Yes	
old_values_json	json/jsonb	No	
new_values_json	json/jsonb	No	
action_note	text	No	
created_at	datetime	Yes	

11.2 reminder_events

****Purpose:**** stores future reminders when available.

Field	Type	Required	Notes
id	UUID / bigint	Yes	
vehicle_id	FK	Yes	
reminder_type	enum	Yes	next_service / next_oil_change / next_puc / insurance_expiry
due_date	date	Yes	
due_odometer	integer	No	useful for oil/service
source_type	text	No	manual/import/system
is_active	boolean	Yes	
created_at	datetime	Yes	
updated_at	datetime	Yes	

12. Recommended enums

customer_type

- retail
- corporate
- insurance
- dealer

jobcard_status

- draft
- open
- in_progress
- waiting_for_parts
- ready_for_review
- ready_for_delivery
- closed
- reopened

- legacy_imported_read_only

part_status

- requested
- approved
- in_stock
- ordered
- partially_received
- dispatched
- used
- rejected
- cancelled

labour_status

- pending
- in_progress
- completed
- approved

payment_status

- unpaid
- partial
- paid

import_row_status

- created
- updated
- merged
- skipped
- failed
- pending_review

import_batch_status

- uploaded
- validated
- imported
- failed

review_status

- pending
- approved_merge
- keep_separate
- rejected

tax_mode

- exclusive
- inclusive

discount_type

- amount

- percent

supplier_order_status

- draft
- ordered
- partial
- received
- cancelled

media_type

- intake_photo
- intake_video
- complaint_audio
- complaint_video
- complaint_photo
- delivery_photo
- work_photo
- signature
- diagnostics_file
- other

13. Key relationship summary

```mermaid

erDiagram

```
CUSTOMERS ||--o{ VEHICLES : owns_currently
VEHICLES ||--o{ VEHICLE_OWNERSHIP_HISTORY : has
CUSTOMERS ||--o{ VEHICLE_OWNERSHIP_HISTORY : appears_in
```

```
CUSTOMERS ||--o{ JOBCARDS : linked_to
VEHICLES ||--o{ JOBCARDS : serviced_by
JOBCARDS ||--|| JOBCARD_SNAPSHOTS : stores
JOBCARDS ||--o{ JOBCARD_MECHANICS : assigned_to
USERS ||--o{ JOBCARD_MECHANICS : works_on
```

```
JOBCARDS ||--o{ JOBCARD_COMPLAINTS : contains
JOBCARD_COMPLAINTS ||--o{ JOBCARD_COMPLAINT_ICONS : tagged_with
COMPLAINT_ICON_MASTER ||--o{ JOBCARD_COMPLAINT_ICONS : selectable
```

```
JOBCARDS ||--o{ JOBCARD_LABOUR : has
JOBCARDS ||--o{ JOBCARD_PARTS : has
```

```
JOBCARDS ||--o{ JOBCARD_MEDIA : stores
VEHICLES ||--o{ DIAGNOSTICS_REPORTS : history
JOBCARDS ||--o{ DIAGNOSTICS_REPORTS : linked
```

```
SUPPLIERS ||--o{ SUPPLIER_ORDERS : receives
SUPPLIER_ORDERS ||--o{ SUPPLIER_ORDER_LINES : contains
```

```
JOB_CARD_PARTS ||--o{ PARTS_DISPATCH_LOGS : dispatched_as
IMPORT_PROFILES ||--o{ IMPORT_BATCHES : used_in
IMPORT_BATCHES ||--o{ IMPORT_BATCH_ROWS : contains
IMPORT_BATCH_ROWS ||--o{ DUPLICATE_REVIEW_QUEUE : creates
DUPLICATE_REVIEW_QUEUE ||--o{ MERGE_LOGS : resolves_into
...
```

---

## ## 14. Deletion and record safety rules

### ### Never hard-delete these when linked to history

- customers
- vehicles
- jobcards
- parts\_master used historically
- labour\_master used historically
- suppliers used in old orders

### ### Preferred behavior

- mark records as inactive
- keep historical references intact
- use merge logs instead of destructive replacement

### ### Imported history

- imported legacy job cards should default to:
  - `legacy\_import\_flag = true`
  - `read\_only\_flag = true`

---

## ## 15. Future-ready fields reserved

These are not active in MVP but should be anticipated:

### ### Zoho integration

Possible future fields already anticipated in jobcards and masters:

- zoho\_sync\_status
- future `zoho\_contact\_id` on customers
- future `zoho\_item\_id` on parts\_master and labour\_master
- future `zoho\_estimate\_id` / `zoho\_invoice\_id` on jobcards

### ### Multi-tenant SaaS later

Every table should be created so that a future `tenant\_id` can be added without redesign.

For MVP, keep schema naming and code structure tenant-ready.

---

## ## 16. Developer notes

### ### Recommended database engine

- \*\*PostgreSQL\*\*

### ### Recommended data types

- UUID or bigint primary keys
- jsonb for import mappings, unknown fields, and audit diffs
- decimal(12,2) for money
- timestamp with timezone for event times where possible

### ### Recommended indexing priorities

- customers.primary\_mobile
- customers.gst\_number
- vehicles.registration\_number\_normalized
- vehicles.vin
- jobcards.jobcard\_number
- jobcards.vehicle\_id
- jobcards.customer\_id
- jobcard\_parts.status
- import\_batches.batch\_code
- duplicate\_review\_queue.review\_status

---

## ## 17. What this schema does not yet define

This document defines the **data model**, not the final SQL migration files. The next technical layer after this should be:

1. exact SQL/Prisma table definitions
2. foreign key constraints
3. indexes
4. API contracts
5. import mapping spec using your legacy CSV files

---

## ## 18. Recommended next document

The best next document after this is:

### ## Legacy Data Mapping Specification

That document should map your uploaded files into these tables exactly:

- Partners.csv -> customers
- Auto object table.csv -> vehicles
- Job cards.csv -> jobcards and snapshots
- Problems.csv -> complaints / complaint presets
- Items.csv -> parts\_master



- Items 1.csv -> labour\_master
- Deadlines.csv -> reminder\_events where usable

That will allow development of the import engine with real data, not assumptions.